Learning Rate Matters:
Vanilla LoRA May Suffice for LLM Fine-tuning

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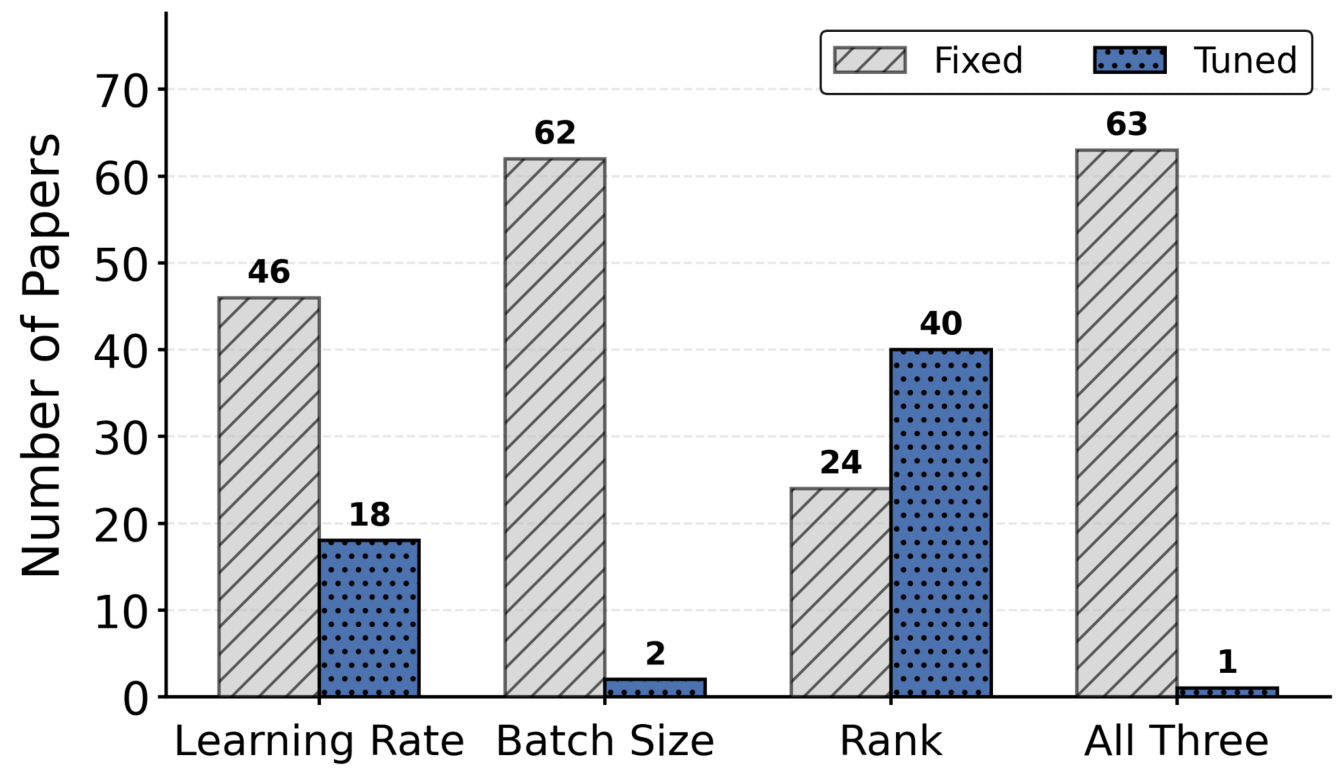
1. What is the problem?

- We conduct a comprehensive audit of advanced LoRA studies to investigate whether their training involved tuning key hyperparameters — **learning rate**, **batch size**, and **LoRA rank**.
- We identify a recurring issue: **the majority of works lack thorough hyperparameter tuning**.
- Among 64 publications collected from top AI conferences and journals, only 1 tunes all three hyperparameters simultaneously, with **up to 70% report results under a fixed learning rate**.



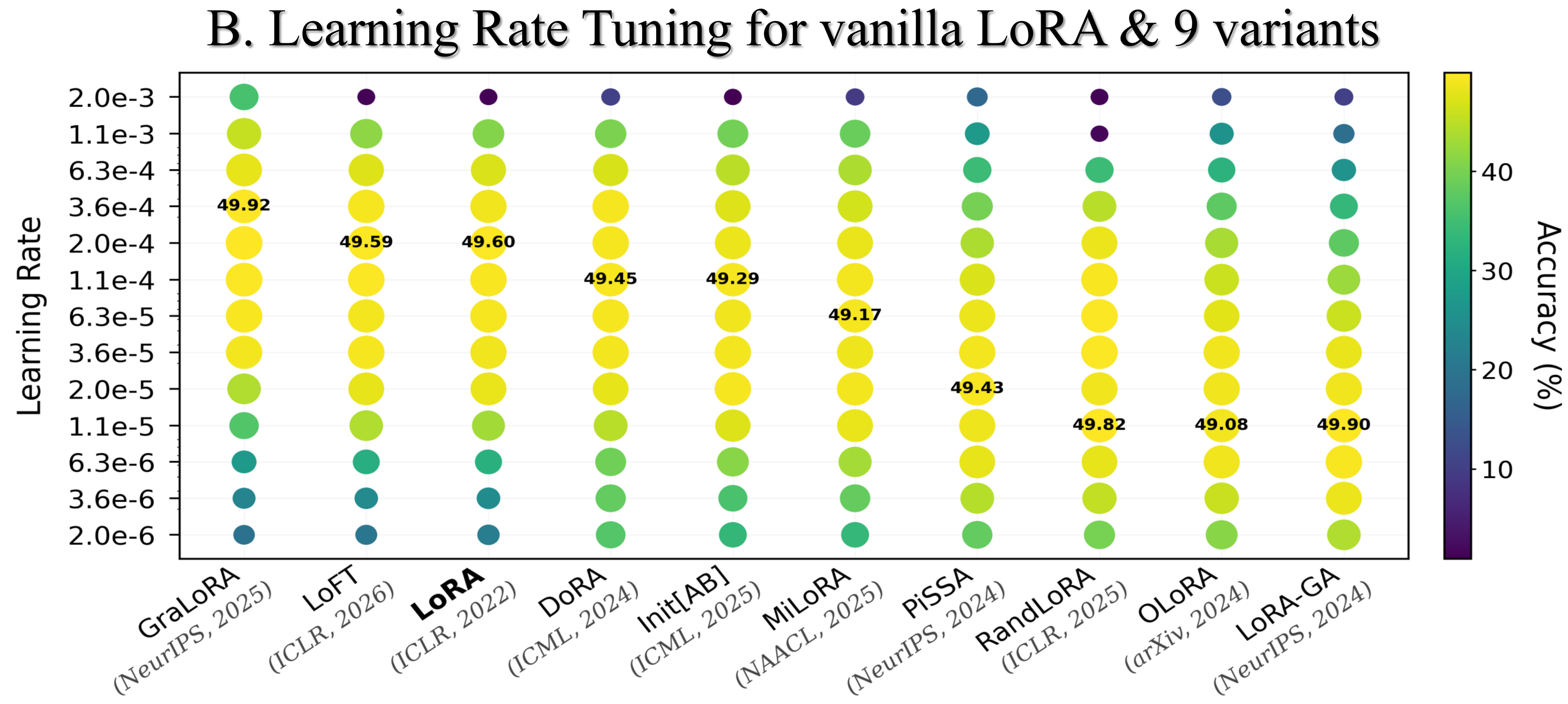
2. Why does it matter?

- Without proper hyperparameter tuning, conclusions may be ungrounded.
- Through extensive fine-tuning experiments across **4 models, 4 task types, and diverse LoRA ranks and training durations**, we demonstrate that different methods require distinct optimal learning rate ranges.
- When configure to their optimal learning rate, all methods yield comparable peak performance, exhibiting no systematic advantage over vanilla LoRA.

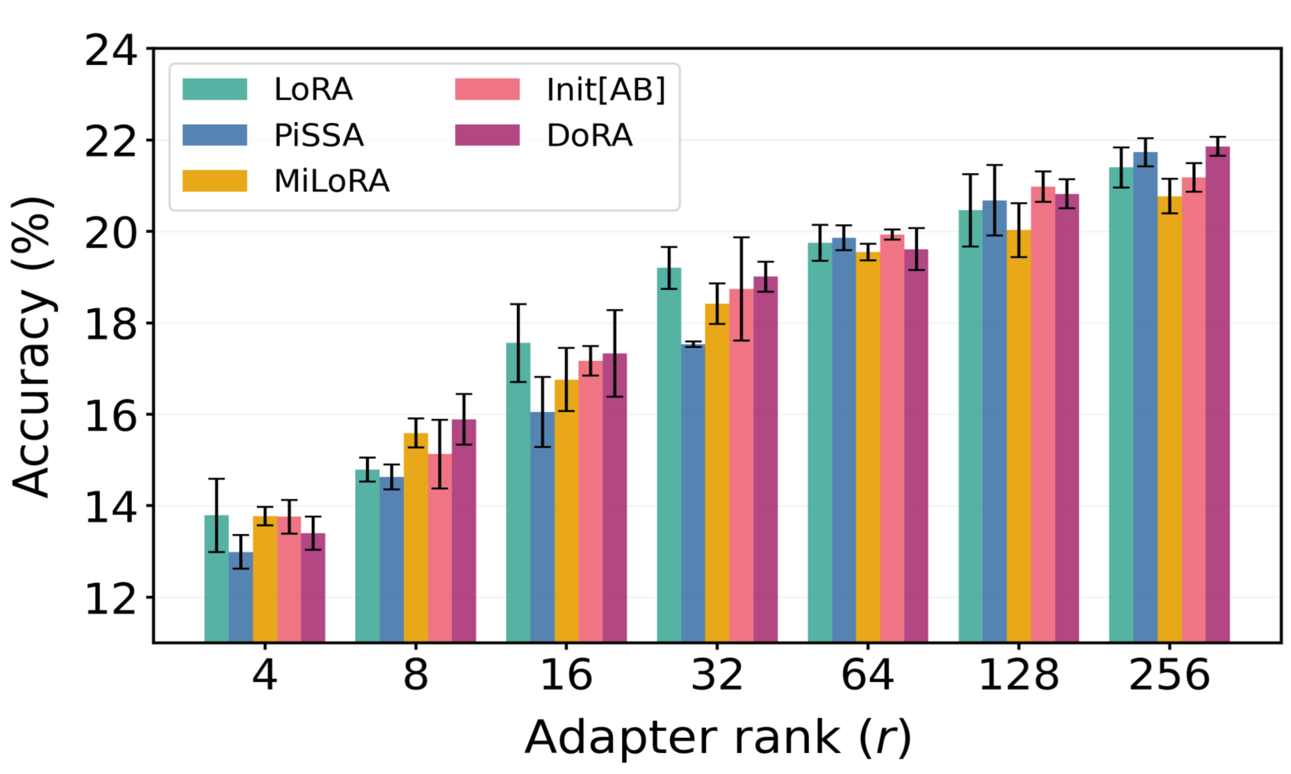
A. Extensive Hyperparameter Tuning Experiments

Model	Task	Rank (r)	Batch Size (B)	Learning Rate (η)
Qwen3-0.6B	Math	8	{16, 64, 128}	1.1247e-5 – 6.3246e-3
		128	64	2.0000e-6 – 2.0000e-3
	Instruction	128	64	2.0000e-6 – 2.0000e-3
Gemma-3-1B	Commonsense	4	64	2.0000e-6 – 2.0000e-2
	Math	{4, 8, 16, 32, 64, 128, 256}	64	1.1247e-5 – 6.3246e-3
		128	{16, 64, 128}	1.1247e-5 – 6.3246e-3
	Code	{4, 8, 16, 32, 64, 128, 256}	64	1.1247e-5 – 6.3246e-3
Llama-2-7B	Math	{8, 32, 128}	{16, 128}	2.0000e-5 – 3.5566e-3
	Code	{8, 32, 128}	{16, 128}	2.0000e-5 – 3.5566e-3
Llama-2-13B	Math	{8, 128}	64	2.0000e-5 – 2.0000e-3

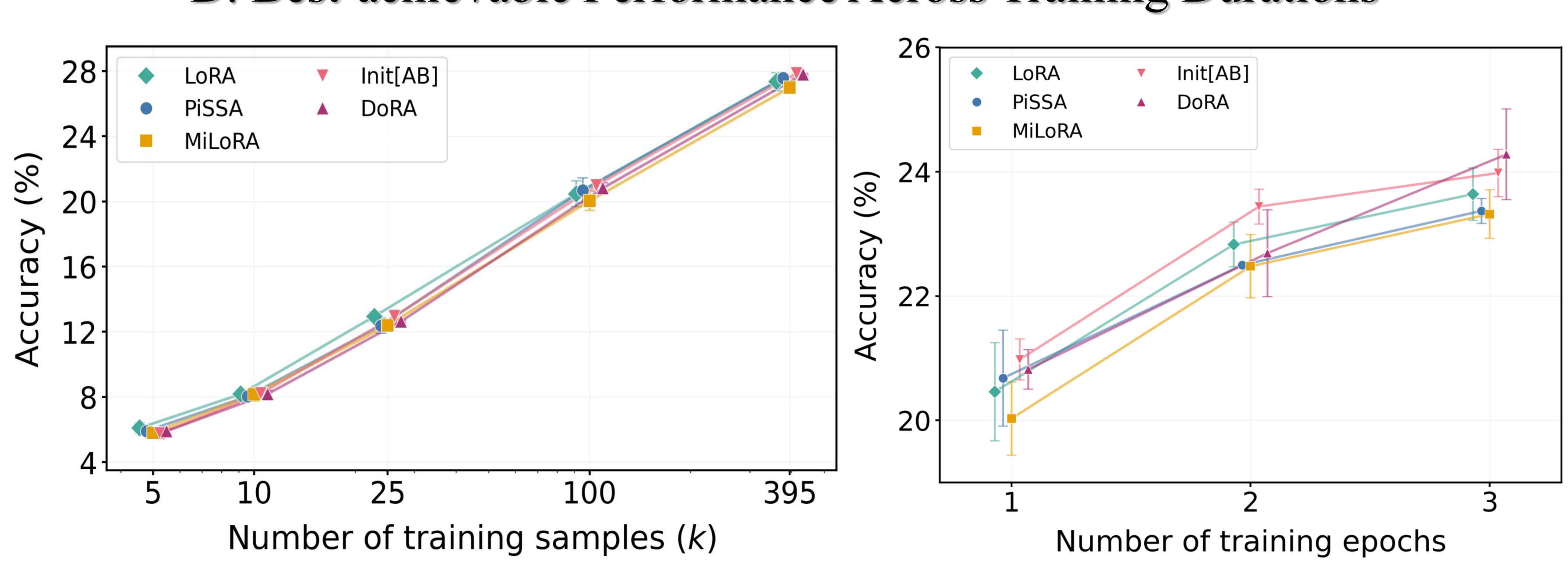
B. Learning Rate Tuning for vanilla LoRA & 9 variants



C. Best-achievable Performance Across Ranks



D. Best-achievable Performance Across Training Durations



3. Which learning rate to use?

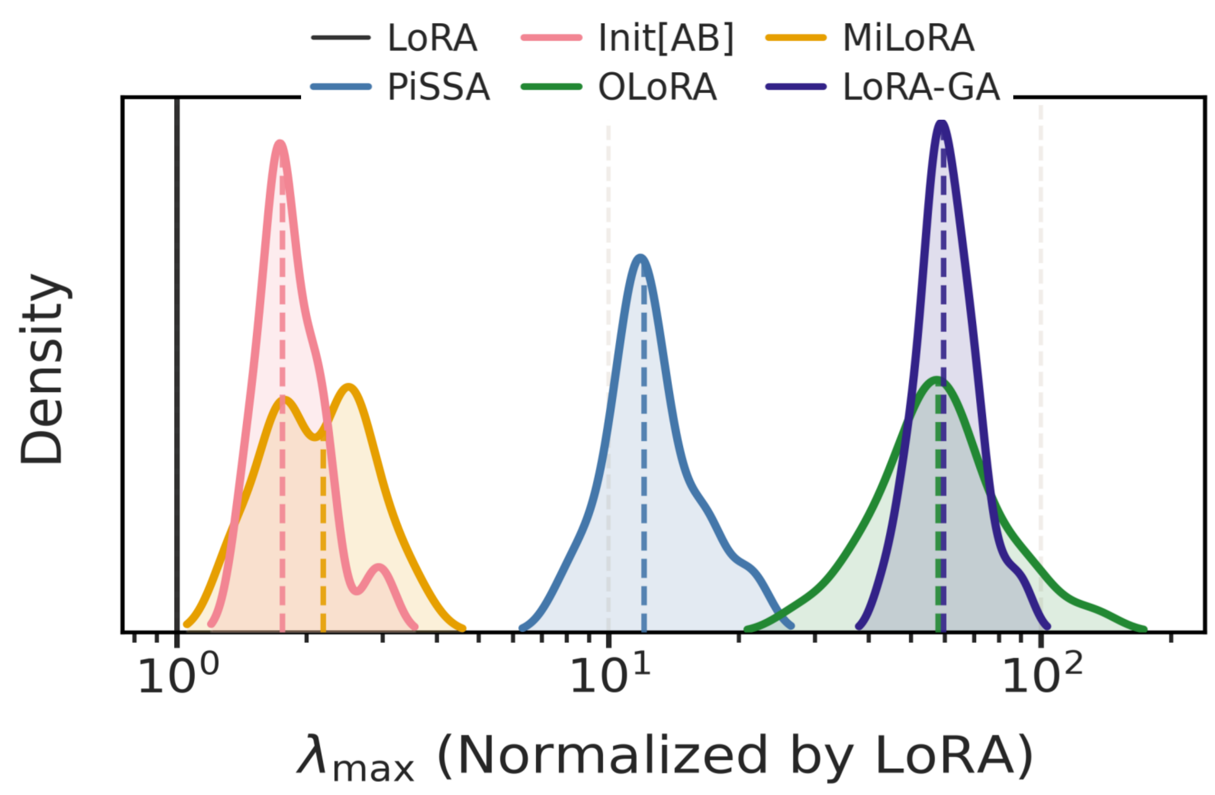
- Different initializations $\boldsymbol{\theta}_0$ induce training trajectories distinct from vanilla LoRA:

$$\{\boldsymbol{\theta}_0^{(i)}\}_{i=1}^{N} \rightarrow \boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \boldsymbol{g}(\theta_t)$$

- Hessian analysis: **optimal LR correlates negatively with the maximum eigenvalue** — consistent with classical theory.

$$\boldsymbol{H}(\boldsymbol{\theta}) \triangleq \nabla^2 \mathcal{L}(\boldsymbol{\theta})$$

$$\eta^* \propto \frac{1}{\lambda_{max}(\boldsymbol{H}(\boldsymbol{\theta}_0))}$$



4. How to tune LoRA methods efficiently?

I. **Prioritize Learning Rate Tuning**

II. **Mind Batch Size Scaling**

III. **Select learning rate based on Hessian**

IV. **Increase LoRA ranks**

V. **Prolong Training Durations**

See paper and project page for more details!